UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 4

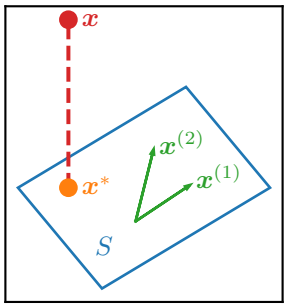
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The previous formula was for projection onto a one-dimensional subspace, and we need to generalize it.
Let S be a d-dimensional subspace. Define a basis $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ for S, thus $S = \text{span}(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(d)})$.
Let's project a vector $\boldsymbol{x} \in \mathbb{R}^n$ onto S:



Since $\boldsymbol{x}^*$ belongs to S, it can be written as:

$$\boldsymbol{x}^* = \alpha_1 \boldsymbol{x}^{(1)} + \alpha_2 \boldsymbol{x}^{(2)} + \cdots + \alpha_d \boldsymbol{x}^{(d)}.$$

From projection, we know:

$$\langle \boldsymbol{x} - \boldsymbol{x}^*, \boldsymbol{y} \rangle = 0, \ \forall \boldsymbol{y} \in S.$$

Let's pick $\boldsymbol{y}$ to be $\boldsymbol{x}^{(j)}$ for $j = 1, \ldots, d$, then

$$\left\langle \boldsymbol{x} - \sum_{i=1}^{d} \alpha_i \boldsymbol{x}^{(i)}, \boldsymbol{x}^{(j)} \right\rangle = 0, \ j = 1, \ldots, d$$

$$\Rightarrow \sum_{i=1}^{d} \alpha_i \langle \boldsymbol{x}^{(i)}, \boldsymbol{x}^{(j)} \rangle = \langle \boldsymbol{x}, \boldsymbol{x}^{(j)} \rangle, \ j = 1, \ldots, d \tag{1}$$

$$\Rightarrow \begin{cases} \alpha_1 \langle \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(1)} \rangle + \alpha_2 \langle \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(1)} \rangle + \cdots + \alpha_d \langle \boldsymbol{x}^{(d)}, \boldsymbol{x}^{(1)} \rangle = \langle \boldsymbol{x}, \boldsymbol{x}^{(1)} \rangle, \\ \quad \vdots \\ \alpha_1 \langle \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(d)} \rangle + \alpha_2 \langle \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(d)} \rangle + \cdots + \alpha_d \langle \boldsymbol{x}^{(d)}, \boldsymbol{x}^{(d)} \rangle = \langle \boldsymbol{x}, \boldsymbol{x}^{(d)} \rangle \end{cases}$$

$\Rightarrow d$ equations and d unknowns: $\alpha_1, \ldots, \alpha_d$.

If the basis set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}\}$ is chosen to be orthonormal, then $\langle \boldsymbol{x}^{(i)}, \boldsymbol{x}^{(j)} \rangle = 0, i \neq j$. Thus we have

$$\text{Equation (1)} \Rightarrow \alpha_j \underbrace{\langle \boldsymbol{x}^{(j)}, \boldsymbol{x}^{(j)} \rangle}_{= \|\boldsymbol{x}^{(j)}\|^2 = 1} = \langle \boldsymbol{x}, \boldsymbol{x}^{(j)} \rangle$$

$$\Rightarrow \Pi_S(\boldsymbol{x}) = \boldsymbol{x}^* = \langle \boldsymbol{x}, \boldsymbol{x}^{(1)} \rangle \boldsymbol{x}^{(1)} + \langle \boldsymbol{x}, \boldsymbol{x}^{(2)} \rangle \boldsymbol{x}^{(2)} + \cdots + \langle \boldsymbol{x}, \boldsymbol{x}^{(d)} \rangle \boldsymbol{x}^{(d)}.$$

Hyperplanes play a big role in convex optimization.

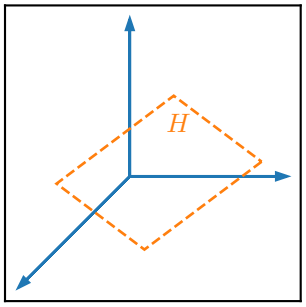
Definition 1 (Hyperplane)**.** A hyperplane H is an $(n-1)$-dimensional affine set that can be written as:

$$H = \left\{ \boldsymbol{z} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}} \boldsymbol{z} = b \right\},$$

where $b \in \mathbb{R}$ is a constant, and $\boldsymbol{a} \in \mathbb{R}^n$ is a non-zero vector.

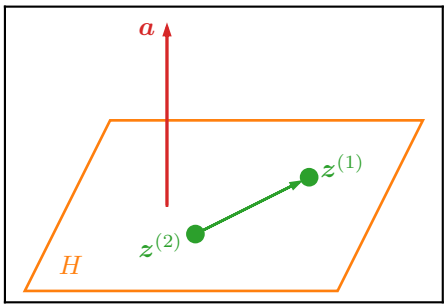
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- In $\mathbb{R}^3$, hyperplane is a 2-dimensional plane:

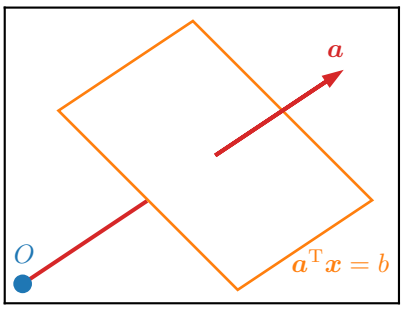


- Pick two vectors $\boldsymbol{z}^{(1)}, \boldsymbol{z}^{(2)}$ in H, then

$$\boldsymbol{a}^{\mathrm{T}}\boldsymbol{z}^{(1)} = b, \boldsymbol{a}^{\mathrm{T}}\boldsymbol{z}^{(2)} = b \Rightarrow \boldsymbol{a}^{\mathrm{T}}\left(\boldsymbol{z}^{(1)} - \boldsymbol{z}^{(2)}\right) = 0$$
$$\Rightarrow \boldsymbol{a} \perp \left(\boldsymbol{z}^{(1)} - \boldsymbol{z}^{(2)}\right).$$



- $\boldsymbol{a}$ is called the *normal vector* of H.

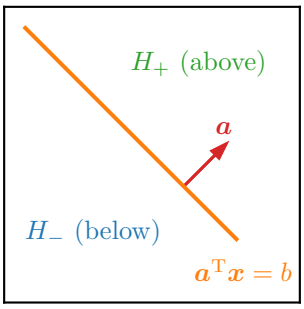


Homework: Find formula for projection onto H (see textbook).

Half-spaces. A hyperplane H divides the space into two regions:

$$H_+ = \left\{\boldsymbol{x} \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \geq b\right\}, \quad H_- = \left\{\boldsymbol{x} \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \leq b\right\}.$$

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- H_+ is in the direction of the normal vector.
- H_- is in the opposite direction.

Later on, you will learn that the gradient of $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}$ is $\boldsymbol{a}$, and moving along gradient increase the function so that $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}$ grows bigger than b. Therefore, H_+ is in that direction. This leads to gradient algorithms to be discusssed later.

Unconstrained optimization.

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \; f(\boldsymbol{x}).$$

- An application of this is least-squares.
- To analyze this, the notion of gradient is important:

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{bmatrix}.$$

(If $n = 1$, the gradient is the same as the derivative.)

Example 1.

- If $f(\boldsymbol{x}) = \sin x_1 + 4x_1x_2 + x_2^2$,

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \cos x_1 + 4x_2 \\ 4x_1 + 2x_2 \end{bmatrix}.$$

- If $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$,

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{x} = 2\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

- If $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2$, the gradient doesn't exist if $\boldsymbol{x} = \boldsymbol{0}$, and if $\boldsymbol{x} \neq \boldsymbol{0}$, we have

$$\nabla f(\boldsymbol{x}) = \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|}.$$

■

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Two important notions in convex optimization:

1. Linear functions: $f(\boldsymbol{x})\colon \mathbb{R}^n \to \mathbb{R}$ is linear if

$$f(\alpha_1 \boldsymbol{x} + \alpha_2 \boldsymbol{y}) = \alpha_1 f(\boldsymbol{x}) + \alpha_2 f(\boldsymbol{y}),\ \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,\ \forall \alpha_2, \alpha_2 \in \mathbb{R}.$$

   If $f(\boldsymbol{x})$ is linear, then there exists $\boldsymbol{a} \in \mathbb{R}^n$ such that $f(\boldsymbol{x}) = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}$.

2. Affine functions: $f(\boldsymbol{x})\colon \mathbb{R}^n \to \mathbb{R}$ is affine if $f(\boldsymbol{x}) - f(\boldsymbol{0})$ is linear, i.e.,

$$\exists \boldsymbol{a} \in \mathbb{R}^n, b \in \mathbb{R},\ \text{s.t. } f(\boldsymbol{x}) = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} + b.$$

   (Note: $f(\boldsymbol{0}) = b$.)
   If $f(\boldsymbol{x}) = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} + b$, then $\nabla f(\boldsymbol{x}) = \boldsymbol{a}$.

Chain rule for gradient. Consider $f\colon \mathbb{R}^m \to \mathbb{R}$ and $\boldsymbol{g}\colon \mathbb{R}^n \to \mathbb{R}^m$. Define $\varphi(\boldsymbol{x}) = f(\boldsymbol{g}(\boldsymbol{x}))$. Let entries of $\boldsymbol{g}(\boldsymbol{x})$ be $[g_1(\boldsymbol{x}), \ldots, g_m(\boldsymbol{x})]^{\mathrm{T}}$. Then,

$$\underbrace{\nabla \varphi(\boldsymbol{x})}_{n \times 1 \text{ vector}} = \underbrace{\begin{bmatrix} \nabla g_1(\boldsymbol{x}) & \cdots & \nabla g_m(\boldsymbol{x}) \end{bmatrix}}_{n \times m \text{ matrix}} \cdot \underbrace{\nabla f(\boldsymbol{z})|_{\boldsymbol{z}=\boldsymbol{g}(\boldsymbol{x})}}_{m \times 1 \text{ vector}}.$$

Each column of $\begin{bmatrix} \nabla g_1(\boldsymbol{x}) & \cdots & \nabla g_m(\boldsymbol{x}) \end{bmatrix}$ is an $n \times 1$ gradient vector.

Example 2. Linear change of coordinates in $\mathbb{R}^n$:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mapsto \quad \begin{bmatrix} 2x_1 + 5x_2 + 1 \\ -x_1 + 6x_2 - 5 \end{bmatrix}.$$

Consider $f(\boldsymbol{x}) = f(x_1, x_2)$ and let $\varphi(\boldsymbol{x}) = f(2x_1 + 5x_2 + 1, -x_1 + 6x_2 - 5)$. By the chain rule, we have

$$\begin{aligned} \nabla \varphi(\boldsymbol{x}) &= \begin{bmatrix} \nabla(2x_1 + 5x_2 + 1) & \nabla(-x_1 + 6x_2 - 5) \end{bmatrix} \cdot \nabla f(\boldsymbol{z})\Big|_{\boldsymbol{z}=\begin{bmatrix} 2x_1 + 5x_2 + 1 \\ -x_1 + 6x_2 - 5 \end{bmatrix}} \\ &= \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix} \cdot \nabla f(\boldsymbol{z})\Big|_{\boldsymbol{z}=\begin{bmatrix} 2x_1 + 5x_2 + 1 \\ -x_1 + 6x_2 - 5 \end{bmatrix}}. \end{aligned}$$

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Taylor series. Given a function $f(\boldsymbol{x})\colon \mathbb{R}^n \to \mathbb{R}$ that is differentiable at a point $\boldsymbol{x}_0 \in \mathbb{R}^n$, it can be approximated by an affine function in a neighborhood of $\boldsymbol{x}_0$:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \varepsilon(\boldsymbol{x}),$$

where the error term $\varepsilon(\boldsymbol{x})$ goes to zero faster than first-order, i.e.,

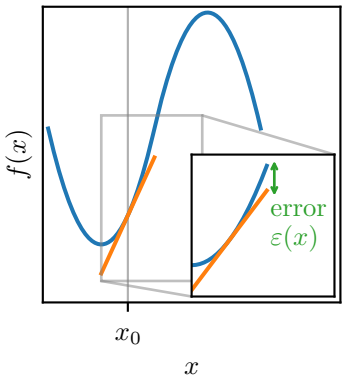
$$\lim_{\boldsymbol{x} \to \boldsymbol{x}_0} \frac{\varepsilon(\boldsymbol{x})}{\underbrace{\|\boldsymbol{x} - \boldsymbol{x}_0\|}_{\text{first order}}} = 0.$$

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This is denoted as

$$f(\boldsymbol{x}) \simeq f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0).$$

Note that the first-order Taylor approximation (i.e., the right-hand side of the above equation) is also an affine function in $\boldsymbol{x}$.

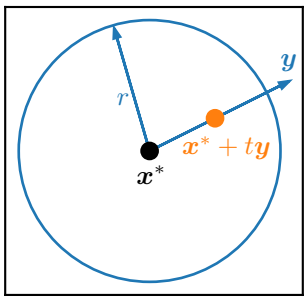


Theorem 1. *If $\boldsymbol{x}^*$ is a local minimum of $f(\boldsymbol{x})$ where $f(\boldsymbol{x})$ is differentiable, then $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.*

Proof. Since $\boldsymbol{x}^*$ is a local minimum, there is a radius $r > 0$ such that

$$f(\boldsymbol{x}^*) < f(\boldsymbol{x}),\ \forall \boldsymbol{x}\colon \underbrace{\|\boldsymbol{x} - \boldsymbol{x}^*\| \le r}_{\text{ball}}.$$

Pick an arbitrary point $\boldsymbol{y} \in \mathbb{R}^n$. For every $t \ge 0$ that is small enough, $\boldsymbol{x}^* + t\boldsymbol{y}$ belongs to the ball:



Thus we have

$$\begin{aligned} & f(\boldsymbol{x}^* + t\boldsymbol{y}) \ge f(\boldsymbol{x}^*),\ \forall t\colon \text{small} \\ \Rightarrow\ & 0 \le \frac{f(\boldsymbol{x}^* + t\boldsymbol{y}) - f(\boldsymbol{x}^*)}{t},\ \forall t\colon \text{small} \\ \Rightarrow\ & 0 \le \lim_{t \downarrow 0} \frac{f(\boldsymbol{x}^* + t\boldsymbol{y}) - f(\boldsymbol{x}^*)}{t}, \end{aligned}$$

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where $\lim_{t\downarrow 0}$ denotes the limit when t goes from the positive direction to zero. Notice that the right-hand side is just

$$\begin{aligned}
\lim_{t\downarrow 0}\frac{f(\boldsymbol{x}^*+t\boldsymbol{y})-f(\boldsymbol{x}^*)}{t} &= \left.\frac{\partial f(\boldsymbol{x}^*+t\boldsymbol{y})}{\partial t}\right|_{t=0} \\
&= \left.\frac{\partial f(x_1^*+ty_1,\ldots,x_n^*+ty_n)}{\partial t}\right|_{t=0} \\
&= \left.\sum_{i=1}^n \frac{\partial(x_i^*+ty_i)}{\partial t}\cdot\frac{\partial f(x_1^*+ty_1,\ldots,x_n^*+ty_n)}{\partial(x_i^*+ty_i)}\right|_{t=0} \qquad (2)\\
&= \left.\sum_{i=1}^n \frac{\partial(x_i^*+ty_i)}{\partial t}\cdot\frac{\partial f(\boldsymbol{z})}{\partial z_i}\right|_{\boldsymbol{z}=\boldsymbol{x}^*+t\boldsymbol{y}\text{ with }t=0} \\
&= \sum_{i=1}^n y_i\frac{\partial f(\boldsymbol{x}^*)}{\partial x_i^*} = \boldsymbol{y}^{\mathrm{T}}\nabla f(\boldsymbol{x}^*),
\end{aligned}$$

where (2) is by the total differential formula, and $\partial f(\boldsymbol{z})/\partial z_i$ is the derivative of f with respect to the i-th argument of $f(\cdot)$. Notice that $\boldsymbol{y}$ is arbitrary. By setting $\boldsymbol{y}$ to $-\nabla f(\boldsymbol{x}^*)$, we have

$$\begin{aligned}
&0 \le -\nabla f(\boldsymbol{x}^*)^{\mathrm{T}}\nabla f(\boldsymbol{x}^*) = -\|\nabla f(\boldsymbol{x}^*)\|^2 \\
\Rightarrow\ &\|\nabla f(\boldsymbol{x}^*)\| = 0 \\
\Rightarrow\ &\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}.
\end{aligned}$$

■

Consider $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, where $f(\boldsymbol{x}) = \|\mathbf{A}\boldsymbol{x}-\boldsymbol{b}\|_2$, with given $\mathbf{A}\in\mathbb{R}^{m\times n}, \boldsymbol{b}\in\mathbb{R}^m$.
This problem is called *least squares* (LS).
To understand this, let's first start with system of linear equations.

Example 3. A system of linear equations with 3 equations and 2 unknowns:

$$\begin{cases} x_1 - x_2 = 1, \\ 2x_1 + 5x_2 = -1, \\ -0.5x_1 + 3.5x_2 = 10 \end{cases}$$

$$\Rightarrow \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 5 \\ -0.5 & 3.5 \end{bmatrix}}_{\mathbf{A}} \cdot \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{bmatrix} 1 \\ -1 \\ 10 \end{bmatrix}}_{\boldsymbol{b}}.$$

(Note: $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ may not have a solution.) ■

Example 4. Fitting a power law to experimental data.
Consider a system with input $\boldsymbol{x}\in\mathbb{R}^n$ and output $y\in\mathbb{R}$. We know that y has a power-law form:

$$y = \alpha x_1^{a_1}\cdots x_n^{a_n}, \qquad (3)$$

where $\alpha, a_1, \ldots, a_n$ are unknown and $\alpha \ge 0$.

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We apply m inputs $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ and measure their corresponding outputs $y^{(1)}, \ldots, y^{(m)}$.
Question: How to find $\alpha, a_1, \ldots, a_n$ from $(\boldsymbol{x}^{(i)}, y^{(i)})$ for $i = 1, \ldots, m$?

$$\text{Equation (3)} \xRightarrow{\log} \underbrace{\log y}_{\tilde{y}} = \underbrace{\log \alpha}_{\tilde{\alpha}} + a_1 \underbrace{\log x_1}_{\tilde{x}_1} + \cdots + a_n \underbrace{\log x_n}_{\tilde{x}_n}$$

$$\Rightarrow \underbrace{\begin{bmatrix} \tilde{y}^{(1)} \\ \vdots \\ \tilde{y}^{(m)} \end{bmatrix}}_{\text{given}} = \underbrace{\begin{bmatrix} 1 & \tilde{x}_1^{(1)} & \cdots & \tilde{x}_n^{(1)} \\ \vdots & \vdots & & \vdots \\ 1 & \tilde{x}_1^{(m)} & \cdots & \tilde{x}_n^{(m)} \end{bmatrix}}_{\text{given}} \underbrace{\begin{bmatrix} \tilde{\alpha} \\ a_1 \\ \vdots \\ a_n \end{bmatrix}}_{\text{unknown}}$$

$\Rightarrow$ We obtained a system of linear equations.

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